

Spin-Resolved VQE-QSE Simulation of Ethylene

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Abstract. We evaluate variational quantum eigensolver (VQE) and quantum subspace expansion (QSE) workflows for planar and rigidly 90°-twisted ethylene using STO-3G CAS(2,2) and CAS(2,4) Hamiltonians. Independent determinant-space CASCI validates the mapped spectra to approximately $10^{-13} E_h$. Ideal UCCSD reproduces all target singlet energies to at most $1.24 \times 10^{-8} E_h$, whereas a shallow hardware-efficient ansatz fails in the enlarged active space. State-matched QSE exactly recovers the selected active-space excitations, but the planar bright $\pi \rightarrow \pi^*$ gap is 14.16 eV rather than the experimental value near 7.6 eV. The historical analysis of Dunning, Hunt, and Goddard explains this as a missing diffuse-state/basis effect rather than a QSE error. On IBM Fez, readout mitigation reduces the planar ground-state bias from 7.53 to 1.74 mE_h and the twisted bias from 12.44 to 6.06 mE_h ; linear zero-noise extrapolation lowers the absolute twisted bias to 2.55 mE_h but overshoots below the exact energy. A post-run audit excludes an observable-layout error and a root-mismatched noisy-QSE result. The study is therefore a validated proof of principle, not yet a quantitative spectroscopic prediction.

Keywords: VQE; quantum subspace expansion; ethylene; excited states; error mitigation; IBM Quantum.

1. Problem and methodology

Ethylene combines a closed-shell planar ground state, a low $\pi\pi^*$ triplet, a bright singlet V state, and strong open-shell character under torsion. It therefore tests state preparation, spin assignment, and spectroscopy rather than energy minimization alone. VQE supplies the reference state [1, 2]; QSE projects the Hamiltonian into a physically motivated excitation subspace [3].

The equilibrium geometry uses $r_{CC} = 1.339 \text{ \AA}$, $r_{CH} = 1.086 \text{ \AA}$, and $\angle HCH = 117.6^\circ$. The twisted structure rotates one CH_2 group rigidly by 90° around C–C. PySCF [4] provides restricted Hartree-Fock STO-3G integrals with the carbon $1s$ core frozen. CAS(2,2) contains π/π^* ; CAS(2,4) adds two frontier spatial orbitals. Jordan-Wigner mapping supports transparent validation, while parity mapping and $\mathbb{Z}_2$ tapering reduce the hardware registers [5].

VQE minimizes $E(\theta) = \langle \Psi(\theta) | \hat{H} | \Psi(\theta) \rangle$. We compare chemically motivated UCCSD [6] with a layered rotation-entangler hardware-efficient ansatz (HEA), using SLSQP in ideal simulation. Particle number and $\langle \hat{S}^2 \rangle$ are audited after optimization; these expectation-value checks are diagnostics rather than exact sector projectors.

For operators $\{O_i\}$, QSE constructs

$$H_{ij} = \langle \Psi_0 | O_i^\dagger \hat{H} O_j | \Psi_0 \rangle, \quad S_{ij} = \langle \Psi_0 | O_i^\dagger O_j | \Psi_0 \rangle, \quad (1)$$

and solves $H\mathbf{c}_k = E_k S\mathbf{c}_k$. Canonical orthogonalization discards overlap eigenvalues below 10^{-8} in ideal calculations [7]. Roots are classified by $\langle \hat{S}^2 \rangle$, transition strength, and state continuity, never by index alone. The recent VQSE preprint of Lamp *et al.* supports this emphasis on reference overlap, operator-pool expressivity, and strict overlap regularization under hardware noise [8].

Noise simulation uses the tapered circuits, a representative FakeManila model, five shot seeds, and 1,024–16,384 shots. Readout mitigation is assignment-matrix inversion followed by nonnegative clipping and renormalization [9]. Linear ZNE uses global folds $s = 1, 3, 5$ [10]. The real-device run used IBM Fez, job: 72 PUBs, 294,912 total shots, and three 4,096-shot repeats. Parameters were optimized classically before submission; this was fixed-circuit energy estimation, not hardware-in-loop VQE or hardware QSE.

2. Validated ideal results

Independent determinant-space CASCI and qubit-space diagonalization agree to about $10^{-13} E_h$, validating integral ordering, core offsets, mapping, and sectors. Table 1 summarizes the state-resolved spectra. At 90° , the global ground state is triplet, only 0.0696 eV below the singlet; optical QSE therefore retains a separate singlet reference.

Table 1: Exact active-space results. q_t/P gives tapered qubits/Pauli terms. Planar gaps are triplet and bright-singlet gaps from S_0 ; twisted gaps are $S_0 - T_0$ and $S_1 - S_0$.

Model	q_t/P	$E_{\text{low}} (E_h)$	Δ_1 (eV)	Δ_2 (eV)
Planar (2,2)	1/15	-77.116592	4.7919	14.1546
Planar (2,4)	3/61	-77.116777	4.7969	14.1596
Twisted (2,2)	2/27	-76.979487	0.0696	6.5802
Twisted (2,4)	5/185	-76.979487	0.0696	6.5235

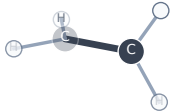
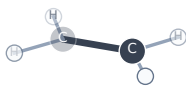
UCCSD errors are at most $1.3 \times 10^{-5} mE_h$ for all four target singlets. HEA is also numerically exact in CAS(2,2), with errors of $4.60 \times 10^{-5} mE_h$ (planar) and $2.31 \times 10^{-5} mE_h$ (twisted), but fails in CAS(2,4): its errors rise to 1.29×10^3 and $1.22 \times 10^2 mE_h$, respectively. Figure 1 shows the tradeoff: the shallow circuit is hardware-friendly but not reliably expressive, whereas mapped UCCSD is accurate but too deep for the enlarged space.

With an identity-plus-single-plus-double pool, ideal QSE reproduces the selected exact gaps to the shown precision. The corrected values are 14.154551 and 14.159602 eV for the planar bright singlet and 6.580248 and 6.523527 eV for the twisted $S_1 - S_0$ transition. The twisted CAS(2,4) value corrects the notebook selector, which originally chose a neighboring 6.557 eV root. This correction illustrates why physical state labels must replace array indices.

The 14.16 eV result is exact only for STO-3G. Dunning *et al.* found the singlet V orbital exceptionally diffuse ($\langle z^2 \rangle \approx 42 a_0^2$ versus $2.7 a_0^2$ for the triplet); diffuse functions lowered their Hartree-Fock gap from 9.10 to 7.42 eV [11]. A state-matched EOM-CCSD audit [12] gives 12.921 (STO-3G), 9.595 (6-31G), 8.245 (6-31+G*), and 8.001 eV (aug-cc-pVDZ) (Figure 2). Thus, active-space enlargement alone cannot supply the missing radial flexibility.

3. Noise and IBM hardware

At 4,096 shots in the retained tapered-circuit simulator, planar errors (mean $\pm$ population SD across five seeds)



C-C = 1.339 Å, C-H = 1.086 Å, H-C-H = 117.6 degrees; only one CH₂ group is rotated.

Figure 1: Notebook-derived models and resources. Left: planar and rigid 90° geometries. Right: UCCSD grows from depth 110/45 two-qubit gates in CAS(2,2) to depth 2,721/708 gates in CAS(2,4), whereas HEA uses depths 17/28 and 6/14 two-qubit gates. All panels are original vector figures regenerated from validated notebook data.

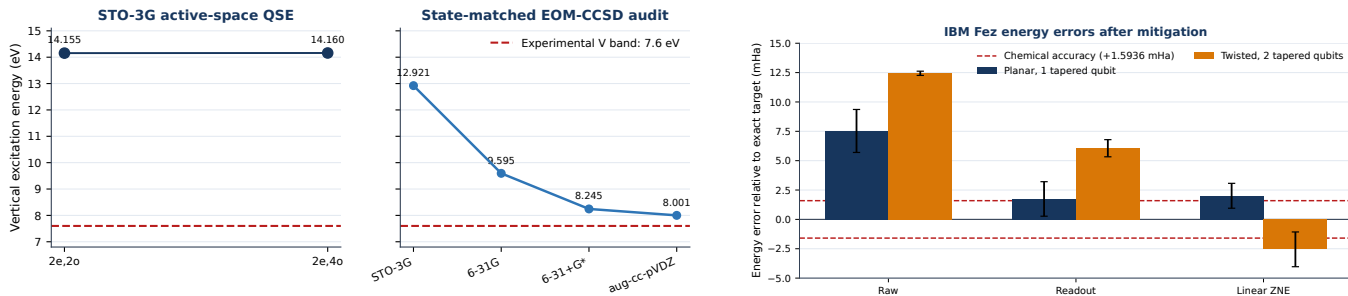


Figure 2: Spectroscopic model error and hardware error are distinct. Left: STO-3G active-space QSE and the state-matched EOM-CCSD basis audit relative to the 7.6 eV band. Right: signed IBM Fez ground-state errors; bars show means and population standard deviations over three repeats. The twisted ZNE result is below the exact energy and therefore nonvariational.

are $26.25 \pm 2.56 mE_h$ raw, $1.38 \pm 2.75 mE_h$ after readout correction, and $1.30 \pm 2.75 mE_h$ after linear ZNE. Twisted errors are 17.02 ± 0.80 , 3.63 ± 0.90 , and $0.39 \pm 1.18 mE_h$. At 16,384 shots, ZNE errors are $1.36 \pm 0.49 mE_h$ and $0.34 \pm 0.26 mE_h$. The simulator is representative, not a Fez calibration model.

Table 2: Signed IBM Fez energy errors in mE_h , mean $\pm$ population SD over three repeats. The planar target is singlet; the twisted hardware target is the global triplet.

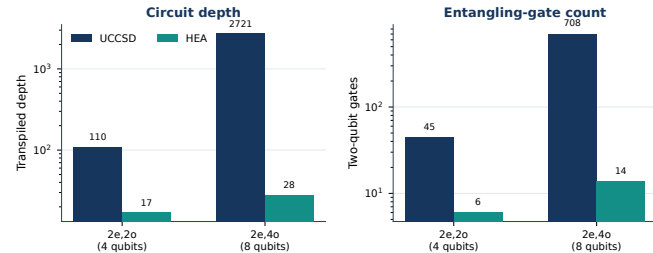
Target	Raw	Readout	Linear ZNE
Planar	$+7.53 \pm 1.83$	$+1.74 \pm 1.47$	$+2.01 \pm 1.06$
Twisted	$+12.44 \pm 0.18$	$+6.06 \pm 0.73$	-2.55 ± 1.48

Readout correction reduces the absolute planar and twisted biases by 76.9% and 51.3%. Linear ZNE reduces the absolute twisted bias by 79.5% but overshoots. None of the hardware means is strictly within the 1 kcal mol⁻¹ = 1.5936 mE_h guide, although the planar readout-corrected mean is close. The three repeats are a descriptive spread from one job, not a confidence interval or drift study.

4. Validation gate and limitations

Excluded outputs. The original untapered four-qubit noise branch paired a transpiled state with logical observables without the final layout; even its noiseless planar check had a spurious +542.66 mE_h error. The sampled noisy-QSE branch subtracted roots 1 and 0, thereby selecting the 4.792 eV triplet and comparing it with the 14.155 eV bright singlet. Neither result is used here.

The retained conclusions are bounded by four limitations. First, STO-3G and the small active spaces cannot provide quantitative V-state spectroscopy. Second, the rigid two-geometry comparison is not a torsional potential-energy surface. Third, expectation-value penalties do not prove exact particle/spin-sector support; variances or symmetry-



preserving constructions are needed. Fourth, raw hardware distributions, physical layouts, calibration snapshots, randomized execution order, and multiple calibration-cycle jobs should be archived for stronger uncertainty claims. A noisy-QSE rerun must preserve complex Hermitian H/S , bootstrap the full pipeline, scan the overlap cutoff, and track state character inside every replicate.

5. Extension and conclusion

A butadiene CAS(2,4) pilot returned an exact energy of $-153.10300349 E_h$ and a UCCSD error of 0.476 mE_h ; single-plus-double QSE recovered the low dark singlet at 7.659 eV, while singles alone did not retrieve it among the corresponding low roots. This demonstrates code-path extensibility, not converged butadiene spectroscopy. For a donor-acceptor study, fragment-aware active orbitals and consistently localized states should precede generalized Mulliken-Hush coupling [13] and Marcus-rate analysis [14]; the exploratory notebook coupling is not reported because it used a difference of dipole norms rather than a common-axis vector projection.

The Hamiltonian construction, ideal UCCSD, and state-matched ideal QSE are validated within their stated models. Symmetry tapering enables the selected fixed circuits on open-access IBM hardware; mitigation reduces but does not eliminate bias. Crucially, the 14.16 eV STO-3G gap is not a prediction of the 7.6 eV ethylene band. The next step is a diffuse, state-balanced treatment followed by state-matched noisy QSE and repeated hardware runs with complete provenance.

Reproducibility statement

The recorded environment was Qiskit 2.5.2, Qiskit Nature 0.8.0, Qiskit Algorithms 0.4.0, Aer 0.17.2, IBM Runtime 0.49.0, PySCF 2.14.0, NumPy 2.1.3, and SciPy 1.16.3.

Seed 42 was used unless a shot-seed sweep is stated. Energy conversion used $1 E_h = 27.211386$ eV. Reported spreads are population standard deviations, not confidence intervals. The executed notebook, LaTeX source, bibliography, and vector-figure generator accompany the report. Full IBM job retrieval may depend on the submitting account and provider retention policy.

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